AMARNATH S. PATEL

Orlando, FL | amarnath.patel@ucf.edu | apatel.co | github.com/jeebuscrossaint | ORCID 0009-0008-9460-082X

EDUCATION

University of Central Florida — **Undergraduate Student** — **4.00 GPA**
B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor — August 2025 – Expected May 2029

Florida Atlantic University — **University coursework via FAU High School, ages 14–18** — **3.66 GPA**
111 credit hours completed between ages 14 and 18, alongside high school — August 2021 – May 2025

RESEARCH EXPERIENCE

Undergraduate Researcher — UCF Astrophotonics Lab, CREOL — August 2025 – Present
Supervisor: Dr. Stephen Eikenberry

- Measuring the wavelength-dependent complex transfer matrix of a photonic lantern by off-axis digital holography, extending Dobias et al., *Opt. Express* **34**(9), 17217 (2026). A photonic lantern couples a multimode fiber to an array of single-mode fibers; using one to feed starlight into single-mode photonic devices requires knowing exactly how light entering each port emerges from the multimode end, in amplitude and phase, at every wavelength of interest.
- Align the interferometric bench and acquire holograms of the multimode output for each input port across the C-band (1525–1575 nm), recovering full amplitude and phase — one row of the transfer matrix per measurement.
- Implemented the phase-retrieval and mode-decomposition analysis: FFT sideband isolation and demodulation, Butterworth low-pass filtering, joint numerical optimization of mode-field diameter, defocus quadratic phase and field position, then decomposition onto the LP basis to recover complex modal amplitude and phase.
- Produced the per-port reconstruction-fidelity characterization for 6- and 7-port lanterns, reaching ≈98% fidelity $|\langle E_{\mathrm{rec}}, E_{\mathrm{sim}}\rangle|^2$ against simulated fields.
- Automated the four-instrument acquisition chain — tunable laser, InGaAs camera, fiber switch and motorized polarization control — so a complete all-port × C-band sweep runs unattended, with polarization optimized in-loop for peak fringe contrast and saturated frames rejected on arrival. A full dataset went from days of manual bench time to a single hands-off run.
- Own the telescope-node software for PolyOculus, an instrument program building photometric observing capability from a networked array of 8 telescopes, aimed at a redshift-drift measurement. Wrote centroid-based autoguiding from the guide camera to INDI mount guide pulses (simulated tracking error 3.01″ ⟶ 0.29″ RMS) and closed-loop piezo steering-mirror control holding a laser spot to 0.2 px, and brought the array's Jetson Nanos up on mainline Linux to replace NVIDIA's end-of-life image, with every node rebuildable from one repository.

Independent Research — CELERIS: computational electromagnetics and metalens design — 2026

- Built a from-scratch solver for Maxwell's equations in periodic subwavelength structures via rigorous coupled-wave analysis / the Fourier modal method: 1D TE/TM multilayer gratings with Li and Liu–Fan inverse-rule factorization, the full 2D-vectorial $P \cdot Q$ formulation for biperiodic nanopillar arrays, and stable Redheffer scattering-matrix layer recursion.
- Validated every layer of the stack against closed-form physics, an independent method, or energy conservation: RCWA versus the transfer-matrix method to 1e-6, multilayer S-matrix to 1e-15, 2D⟶ 1D degenerate reduction to 5e-12, Σ(diffraction efficiency) = 1.000000, and diffraction-limited designed focal spots at FWHM = $\lambda f/D$. Independently cross-validated against the grcwa and Stanford S⁴ solvers (≈ 1e-7) and reproduces canonical published metalenses (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
- Implemented the physical design and analysis capability standalone RCWA kernels lack: Sellmeier and tabulated n, k dispersion, meta-atom library sweeps with automatic etch-depth selection for full-2π phase coverage, gradient-based (Adam) inverse design, and Pancharatnam–Berry geometric-phase and polarization-multiplexed lenses exploiting form birefringence. Analysis covers Strehl ratio, Zernike wavefront, MTF, chromatic focal shift, through-focus caustic and fabrication-tolerance Monte Carlo, with fabrication-ready GDSII export.
- Accelerated Rayleigh–Sommerfeld far-field propagation as a CUDA kernel, benchmarked against the project's own optimized 16-core CPU path: 4.8–5.8× at 92k pillars, agreeing to machine precision. C++23/CUDA with Python bindings. *Sole-author manuscript in preparation (Computer Physics Communications).*

Undergraduate Research Assistant — UCF Physics Department — March 2026 – Present
Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Research on automated assessment generation for introductory physics instruction: built ESTELA, which produces multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, and an agentic generation–evaluation pipeline that iteratively drafts and validates new problems against instructor criteria.
- Materials authored for the project are in use by external adopters following a faculty workshop. Manuscript in preparation.

Undergraduate Researcher — FAU Grant-Funded AI Safety Research Project — January 2024 – March 2025
Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10 statistical coherence and detection measures (GPT-2 perplexity, RoBERTa, BERT next-sentence prediction, LSA, NLI, burstiness) on a 255,000-passage corpus; identified GPT-2 perplexity as the strongest discriminator (3.35× separation, 17.5 versus 58.5) and showed BERT NSP to be uninformative for the task.
- Built four adversarial evasion pipelines (iterative rewriting, tree-search decoding, list-branching, reinforcement learning) over 3,125 generated sequences and characterized a consistent quality–evasion tradeoff. Grant-funded with HPC access.

PRESENTATIONS

Batch Processing for Automated Grading via Azure OpenAI — June 2026
University of Central Florida, Downtown Campus. ESTELA project, Physics Education Research.

Coherence and Detection Approaches for Identifying AI-Generated Text — March 2025
Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

TECHNICAL REPORTS

Coherence and Detection Approaches for Identifying AI-Generated Text — 2025

Empirical study of AI-text detection and adversarial evasion: benchmarked 10 coherence and detection methods and built four evasion pipelines, characterizing a consistent quality–evasion tradeoff. Grant-funded, Florida Atlantic University.

TECHNICAL SKILLS

Instrumentation: interferometric alignment, GPIB/VISA and RS-232 instrument control, INDI, GigE Vision camera acquisition, tunable IR laser control, motorized polarization control, fiber coupling, FITS and GDSII data formats

Scientific computing: FFT and signal processing, phase retrieval and modal decomposition, electromagnetic simulation (RCWA), gradient-based optimization, Monte-Carlo tolerance analysis, GPU/CUDA acceleration, NumPy, SciPy

Programming: C/C++ (C++23), CUDA, Python, Rust, Go, shell (Fish, Bash), LaTeX/Typst

Tools: Linux, Git, CMake, Docker, Qt/PySide6

RELEVANT COURSEWORK

University of Central Florida — *Physics:* Geometric Optics & Lab, Modern Physics, Quantum Information Processing, Mathematical Methods for Physics, Independent Research [PHY 4912], Electricity & Magnetism I (in progress). *Mathematics:* Applied Linear Algebra, Complex Analysis, Linear Algebra (proof-based, in progress), Partial Differential Equations (in progress). *Computer science:* C Programming, Discrete Structures, Object-Oriented Programming.

Florida Atlantic University — General Physics I and II (honors), Calculus I–III (honors), Ordinary Differential Equations, Elementary Matrix Algebra, Data Structures & Algorithms, Computer Architecture, Computer Logic Design, Deep Learning, C++ Programming.

HONORS & AWARDS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition)	2025
Lockheed Martin Award, "Highest Level of Engineering Excellence," AEV Competition	2024
2nd Place in Division, Advanced Experimental Vehicles Competition	2024
1st Place, Night Hacks Hackathon — UniUtils	2023

SERVICE & OUTREACH

Secretary, National Society of Black Physicists — UCF chapter	2025 – Present
Teaching Assistant, Calculus — office hours, exam review and grading for 70 undergraduates	2024 – 2025
Founder and President, Coding Club, FAU High School — now an official Hack Club chapter	2023 – 2025